

Maker Buoy Code Arduino IDE Instructions

V2.1

Step 1. Download latest version of Arduino IDE from: <https://www.arduino.cc/en/software/>

Step 2. Open Buoy Code in IDE

Extract supplied example Arduino Sketch to Documents folder. Double-click to open in Arduino IDE. You'll see an error that it needs to be saved in a folder. IDE will do it automatically.

Step 3. Add Board Support

Go To: Tools->Board->Board Manager. Type "Arduino SAMD" in search bar. Install the latest version, e.g. 1.8.14, if not included in Arduino installation.

Next open File->Preferences->Settings. Where it says "Additional Boards Manager URLs" enter the following and hit OK: https://adafruit.github.io/arduino-board-index/package_adafruit_index.json

Go To: Tools->Boards-Board Manager again. Enter 'Adafruit SAMD' in the search bar. Install the latest version, e.g. 1.7.17.

Step 4. Edit variant.cpp File in Adafruit Folder

Close Arduino IDE. Go to:

C:\Users\<username>\AppData\Local\Arduino15\packages\adafruit\hardware\samd\X.X.X\variants\feather_m0

You may need to show hidden files to see this folder. If you can't find this folder, search for "feather_m0" to find the appropriate folder.

Inside the folder, look for the variant.cpp file. Edit this file in notepad. Go to bottom of the file and comment out the last entry as show below. This is required to compiled code that accesses 2nd serial port.

```
//void SERCOM5_Handler()
//{
//  Serial5.IrqHandler();
//}
```

If your program won't compile, it's most likely the board manager has been upgraded and you forgot to make this change. It must be done for any update to the Adafruit SAMD board manager.

Step 5. Extract Libraries

Find the folder under Documents\Arduino\libraries. If it the "libraries" folder doesn't exists, create it first. Extract the libraries provided and place in that folder. There should be 3 folders in the .ZIP file. It

may take a few min for this to extract.

CAUTION: You may update all libraries EXCEPT Adafruit_VC0706... which has been specially modified to support additional camera commands.

Step 6. Verify Compile

Reopen the Arduino IDE.

Open Buoy Code file again and under Tools->Board select Adafruit SAMD Boards and “Adafruit Feather M0” from the list of processors. Click Verify/Compile to verify you can compile without errors. Assuming no errors, you’re ready to install on the board.

Step 6. Upload Code to Board

Plug the buoy mainboard into your computer via a data micro-USB cable. Next to the USB connector on the Feather M0 board is a small reset button. **Double-click the button** so that you see the status LED increasing and decreasing in intensity. Under Tools->Port select the COM port of the Feather M0. Now go to Sketch->Upload. The code will be uploaded to the board after a few minutes. The IDE window will show “Verify Successful” when complete.

CONGRATULATIONS YOU'RE DONE!